

SIMULATING AND ANALYSING THE MICROVIBRATION SIGNATURE OF REACTION WHEELS FOR FUTURE NON-INTRUSIVE HEALTH MONITORING METHODS

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ABSTRACT

This paper will present an approach to simulate and analyse the microvibration signature of reaction wheels with the goal to improve the understanding of microvibration measurements performed with such mechanisms for the future development of non-intrusive health monitoring systems for space mechanisms.

At first the background of reaction wheel related microvibrations will be discussed, including a typical microvibration signature of a reaction wheel which will be presented. The relevant signal components in measurement data will be identified and described.

Based on this, a method will be presented to simulate the microvibration signature of the reaction wheels. The model approach as well as the fundamental equations and the analysis strategy will be discussed. Both, measurement and simulation results will be compared briefly. The individual effects, which can be described by the model, will be outlined.

The paper concludes with some conceptual ideas on how the modelling and analysis methodology can be applied for an autonomous non-intrusive health monitoring system for future spacecraft mechanisms.

1. INTRODUCTION

The majority of current spacecraft applies ball bearing suspended reaction wheels (RW's) for attitude control and stabilisation. Microvibrations (i.e. disturbing vibrations, which are typically very low in amplitude and widely spread over the frequency spectrum) are emitted due to operation of the RW's. On one hand, such vibrations should be prevented, damped or isolated as much as possible to reduce the disturbance of vibration sensitive payloads like e.g. Earth observation imagers, space telescopes and satellite optical interlink modules, and allow for maximum performance.

On the other hand, these vibrations are caused by defined mechanical and electromagnetic processes inside the reaction wheel assembly to large extent. They carry comprehensive information on the actual health state of the mechanism, such that they could be utilised for a non-intrusive health monitoring system. For large turbines and pumps, e.g. in power plants, this

is a state-of-the-art method to monitor the wear and damage of critical components like the bearings.

It might also be applied for an autonomously operating health monitoring system for quasi-permanently operative space mechanisms like RW or cryo coolers.

As a preparatory step, a study was performed to identify the particular signal components to be observed by such a system, to extract them from a given measurement data set and to define and interpret parameters, which might indicate the health state of the mechanism concerned. In the course of the study a model was developed which is capable of simulating the microvibration signature of a reaction wheel. This provided an enhanced understanding of the vibration components and can be used as a reference when analysing wheel performance. The model can be a basis for the development of an operation health monitoring system.

2. REACTION WHEEL INDUCED MICROVIBRATIONS

2.1. General

For better understanding of the microvibrations generated in reaction wheels their measurable signatures may be analysed. Typically, such signatures of the microvibration behaviour of a rotating mechanism are derived from axial and radial accelerations or forces acquired during spinning up or running down the wheel, for example on a so called dynamometric table. An overall signature can then be displayed in form of waterfall or contour plots (spectral component vs. frequency and wheel speed, or time). This is a common approach and input for space system microvibration assessment [1].

2.2. Signatures of RW Microvibrations

A typical example of a microvibration signature of a reaction wheel shall be used to present the individual signal components emitted during operation of reaction wheels in relation to the corresponding physical processes. Fig. 1 shows the waterfall plot of the radial force during spin-up of a reaction wheel.

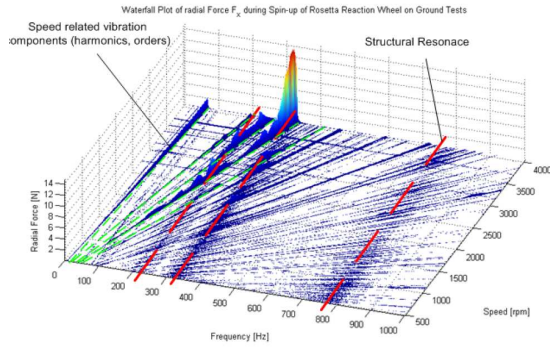


Figure 1. Microvibration signature of a reaction wheel (ROSETTA spacecraft RW, on-ground measurement; courtesy Astrium & ESA).

The radial force has been measured during RW spin-up from 500 revolutions per minute (rpm) to 4000 rpm. Above plot depicts the radial force magnitude (z-axis) as a function of frequency (x-axis) and wheel speed (y-axis).

It can be seen that numerous signal components are aligned with rays from the origin of the diagram (hence, wheel speed related microvibration features). Other components manifest themselves as features parallel to the wheel speed axis (i.e., not changing with wheel speed) at around 220 Hz, 320 Hz and 780 Hz.

The first set of components describe harmonic orders of the wheel speed. They are proportional to the wheel speed concerning their frequency, and they represent the actual excitation sources of the microvibrations. The second set indicates the structural behaviour, in particular the structural resonances of the reaction wheel tested, which tend to amplify the excitation forces in certain frequency bands.

2.3. Microvibration Sources in Reaction Wheels - Harmonic Orders

Due to the very good correlation of the frequency of excitation forces and the wheel speed, it can be concluded that these are due to deterministic mechanical and electromagnetic processes inside the reaction wheel assembly. The corresponding effects are governed by several design parameters and can be empirically estimated from measured signatures.

The main excitation effects are due to the rotor unbalance, the bearings of the reaction wheel and due to the motor/ motor controller. The effects are described well in literature [2], [3], [4] and shall only be briefly discussed at this occasion.

Primary Excitation Features:

The basic excitations of the system are namely static and dynamic unbalance, (ball) bearing internal mechanics and motor operation (i.e. related to number of pole pairs, commutation, etc.). The bearing related effects are among others caused by variable compliance of the bearing as it rotates, as well as

certain imperfections of the bearings, which can be classified as:

- Inner and outer race imperfection,
- Ball imperfection,
- Cage rotation related disturbances.

In this context, the term “imperfection” shall cover local defects on the raceway or the rolling elements like particles or local wear as well as “global” shape errors like race waviness, e.g. introduced by manufacturing. All these effects cause excitations at distinct harmonic orders which are basically caused by the geometry of the bearing. In [4], [5], [6] the corresponding formulae are given to calculate the disturbance frequencies

Secondary Excitation Features:

According to simplified models, the effects and frequencies associated with ball bearings are rather well defined. In practice however, bearing related vibrations not only appear at the above-mentioned fundamental harmonics, but in a more complex way, which can be considered as disturbances derived from the primary ball bearing defect frequencies. Basically, they can be linked to two aspects, as illustrated in Fig. 2.

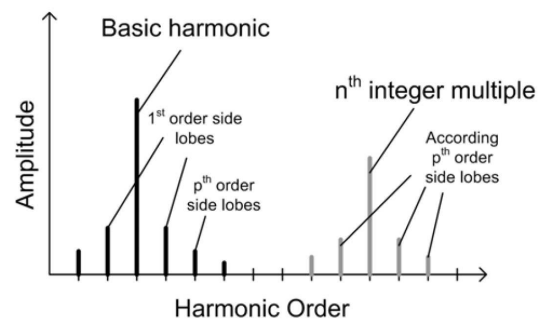


Figure 2. Secondary excitation harmonics

Secondary features are either integer multiples of the basic harmonic or side band components, which may be actually caused by modulations of the corresponding basic harmonic. Some of the side band components can be explained by outer raceway waviness (or ball waviness). Each multiple of the basic harmonic in itself can have side band components. The magnitude of secondary excitation features usually diminishes rapidly with increasing harmonic order, such that they might not be recognisable in particular measurement data. However, as they might interfere with structural resonances, they can considerably contribute to the emitted microvibrations. Tab. 1 provides typical derived bearing harmonics (based on the equations given in [4] and [5]).

In addition to this, investigations on the measurement data presented in figure 1 have unveiled that specific disturbance effects can be identified while considering only half the number of the rolling elements (instead of the full complement). Such effects have been named “Half ball complement related bearing harmonics.”

Imperfection	Main Lobe Harmonic	Side Lobe Harmonics
Outer Raceway Imperfection (ORI)	$k \cdot H_{ORI}$	$k \cdot H_{ORI} \pm p$ $k \cdot H_{ORI} \pm p \cdot H_{CR}$
Inner Raceway Imperfection (IRI)	$k \cdot H_{IRI}$	$k \cdot H_{IRI} \pm p$ $k \cdot H_{IRI} \pm p \cdot (1 - H_{CR})$
Ball Imperfection (BI)	$k \cdot H_{BI}$	$k \cdot H_{BI} \pm p \cdot H_{CR}$ $k \cdot H_{BI} \pm p \cdot (1 - H_{CR})$ $k \cdot H_{BI} \pm p \cdot H_{BS}$

(k, p = 1,2,3,..., CR = Cage Rotation)

Table 1. Typical derived bearing harmonics

Taking this aspect into account, most of the excitation components in a given spectrum can be identified. Nevertheless, various non-deterministic effects (e.g. due to lubrication or other non-linear effects) contribute to the emitted microvibrations, too. Due to their considerably lower signal power, they can be considered not relevant in this context however.

3. REACTION WHEEL MICROVIBRATION DISTURBANCE MODEL

3.1. Model Approach

To improve the understanding on the background of the signatures, hence for better interpretation of measurement results, regarding the microvibration behaviour of reaction wheels, a generic model was developed in the course of the study. As an example, the signature has been simulated by modelling the spin-up of a selected wheel in a defined wheel speed range. The model presented here is intended to be a basic, but versatile model to evaluate disturbances arising from the operation of reaction wheels. Throughout this paper, it will be referred to as Reaction wheel microvibration Disturbance Model (RDM). In some terms, the model relies on earlier work in this area [7], [8], [9], but also covers additional aspects not considered at that level of detail, yet. In particular, the following objectives are covered by this model:

- Simulation of disturbance forces and torques
- Output of reaction forces and torques in time and frequency/order domains
- Consideration of reaction wheel and interface structural dynamics
- Parameterisation using modal characteristics
- High flexibility with respect to model and simulation parameters like simulation duration, wheel speed range, sampling frequency, spectral resolution, etc.

One of the core of the RDM is to define a complex force-to-force transfer function for the wheel structure and the interface structure based on modal superposition [10] applying condensed modal parameters, rather than developing and solving the equations of motion of the whole system like presented in [7]. This approach may also be referred to as transmissibility approach which is described in [11].

By utilising this approach different structural resonances can be considered and additional modes can be added if deemed necessary.

The reaction forces and torques can be evaluated in the frequency or order domain:

$$|F_{out}(f, \Omega)| = \Gamma(f, \Omega) \cdot |F_{in}(f, \Omega)|, \quad (1)$$

with Γ being defined as the transmissibility in frequency or order domain depending on the excitation frequency f and the rotational speed of the wheel Ω and $|F_{out}|$ and $|F_{in}|$ as the magnitudes of the output and input forces/torques. As in a reaction wheel the excitation is most significantly caused by unbalance and bearing related effects, these features will be taken into account for modelling. The excitation is considered as a sum of the corresponding harmonics (sum of sinusoids) with the amplitudes changing with the instantaneous wheel speed. The resulting forces/torques are then evaluated in the time, frequency and order domain and can be presented in waterfall or contour plots. Furthermore, the amplitude-speed relation of certain harmonic orders can be plotted versus given measurement data to allow for model correlation.

3.2. Model Structure and Equations

The RDM basic configuration is illustrated depicted in Fig. 3. It includes the dynamical behaviour of the reaction wheel structure as well as the interface structure. Both are represented as masses connected via modal springs and dampers which represent the condensed dynamic behaviour of the respective structure. The excitation forces and torques, modelled as harmonic disturbances act on the shaft of the wheel and are then transferred to the interface structure. The output or reaction forces can be considered to act on the boundary to any other adjacent structure of a spacecraft or the test facility.

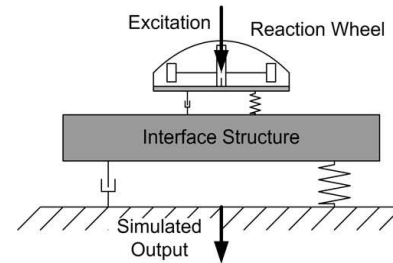


Figure 3. Schematic illustration of the RDM

First, the structural behaviour in terms of the transmissibility of the system is calculated. After that, the excitation (input) forces are calculated for the required number of harmonic orders. By default, 30 primary and secondary excitation harmonics are regarded by the model. Additional harmonics may be added as necessary.

Transfer function

The main idea for estimating the transmissibility (or, generally speaking, the transfer function of the system) is based on the principle of modal superposition.

Modal superposition assumes that the dynamic behaviour of a complex system can be modelled as a linear combination of the (less complex) dynamic behaviour of several elementary systems [10]. Following this, the mechanical system of a reaction wheel mounted on a certain interface structure can be considered as a system of several modal masses connected by a linear spring and a proportional damper in each spatial direction. The RDM therefore can feature a maximum of 9 modal oscillators per direction (4 wheel modes [3] and max. 5 base modes). For each of the modal subsystems a complex transfer function $H(f)$ can be calculated using the Fourier transformation of the underlying equation of motion. Introducing the eigenfrequency f_0 and the damping ratio ξ this can be written as

$$H(f) = \frac{1 + 2 \cdot i \cdot \xi \cdot \frac{f}{f_0}}{\left(1 - \left(\frac{f}{f_0}\right)^2\right) + 2 \cdot i \cdot \xi \cdot \frac{f}{f_0}}. \quad (2)$$

The transfer functions are superposed and weighted according to the respective mode shape functions, which is effectively regarded by mode shape factors or modal participation factors (q). The overall transmissibility is then calculated by

$$\Gamma(f) = \left| \sum_{m=1}^M q_m \cdot H_m(f) \right|, \quad (3)$$

where M denotes the number of modes included and q_m defines the corresponding modal participation factor. For the simulation of the signal in the time domain, the phase can be derived from the real and imaginary parts of the complex modal superposition.

The transmissibility approach is actually only valid for linear systems. Hence, the used parameters need to be constant over time. However, for gyroscopic effects of the spinning wheel the related nutation and precession modes and corresponding damping ratios become functions of the wheel speed, which might be changing during the test or simulation. To meet the requirement for applying the transmissibility approach nevertheless, the previously defined continuous functions are transformed into discrete arrays. For this purpose, wheel speed and frequency (order) are provided as vectors. Between two samples of these vectors, the speed is considered constant. The transfer function might be applied to each speed step without violating the requirement mentioned above, since the parameters are not varying. Hence, to regard the gyroscopic effect as function of the wheel speed, the transfer function is recalculated for each speed step. As result, the transfer function used in the RDM becomes a matrix of discrete

complex quantities and the gyroscopic effect can be regarded without violating the requirement of a linear system. The elements of the complex transfer function matrices for one mode in the frequency and order domains may be therefore calculated by

$$[H_{r,c}^{freq}] = \frac{1 + 2 \cdot i \cdot \xi(\Omega(c)) \cdot \frac{f(r)}{f_0(\Omega(c))}}{\left(1 - \left(\frac{f(r)}{f_0(\Omega(c))}\right)^2\right) + 2 \cdot i \cdot \xi(\Omega(c)) \cdot \frac{f(r)}{f_0(\Omega(c))}}, \quad (4)$$

$$[H_{r,c}^{ord}] = \frac{1 + 2 \cdot i \cdot \xi(\Omega(c)) \cdot \frac{h(r) \cdot \Omega(c)}{f_0(\Omega(c))}}{\left(1 - \left(\frac{h(r) \cdot \Omega(c)}{f_0(\Omega(c))}\right)^2\right) + 2 \cdot i \cdot \xi(\Omega(c)) \cdot \frac{h(r) \cdot \Omega(c)}{f_0(\Omega(c))}} \quad (5)$$

with r and c being the respective row and column/vector indices and h being the order vector. The formulae for the wheel speed depending eigenfrequencies and damping ratios can be derived from [3] and [7].

Excitation Forces

Once the transmissibility matrices of the entire system are derived, the excitation forces/torques need to be defined. Within the RDM, the according definition is also given in frequency/order domain matrices such that the output can be calculated by element-wise multiplication of the input force matrix with the transfer function matrix.

The actual simulation of the input or excitation force/torque is based on the assumption that all disturbance sources discussed earlier can be considered as sinusoids, which can be superposed either in the time, frequency or order domain. Each disturbance source k can be characterised by its amplitude, its harmonic order h and its initial phase angle φ . It should be recalled that a certain source k also causes D disturbances at multiples of it and in related side bands. The definition of a specific disturbance force (index d), say in radial direction, might be given in the time domain by

$$F_{in,k,d}(\Omega, t) = \hat{F}_{in,k,d}(\Omega) \cdot \sin(h_{k,d} \cdot \Omega \cdot t + \varphi_{k,d}). \quad (6)$$

To model the amplitude of a certain harmonic as function of the wheel speed the equation given in [7] can be generalised in the form of

$$\hat{F}_{in,k,d}(\Omega) = a_{k,d} \cdot \Omega^{b_{k,d}} \text{ (forces) and} \quad (7a)$$

$$\hat{T}_{in,k,d}(\Omega) = c_{k,d} \cdot \Omega^{d_{k,d}} \text{ (torques)} \quad (7b)$$

With a, b, c, d being amplitude coefficients to characterise each (k, d) harmonic excitation.

By summing up over all indices k and d the total excitation force/torque can be obtained in the time domain as function of the wheel speed by

$$F_{in,k,d}(\Omega, t) = \sum_{k=1}^K \sum_{d=1}^D F_{in,k,d}(\Omega, t). \quad (8)$$

The required matrices in frequency and order domain are derived via Fourier transforms.

3.3. Exemplary RDM Results

To show the performance of the RDM, the achieved results in the frequency domain are presented and compared to measurement data obtained with a specific reaction wheel on the test bed.

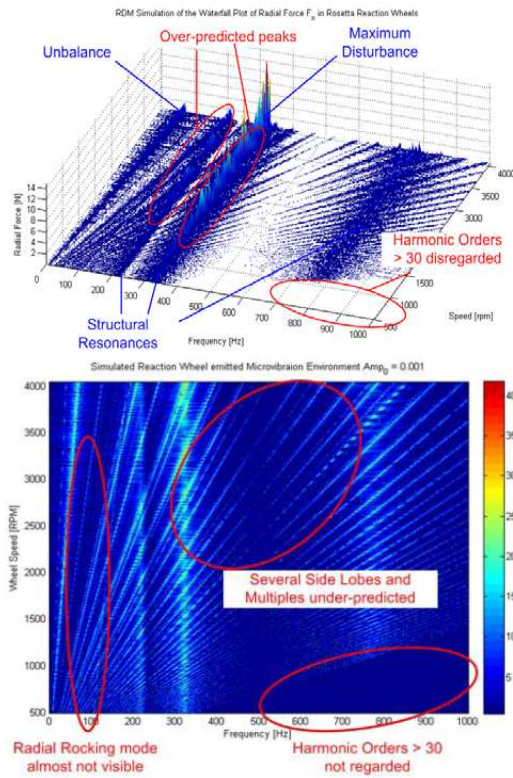


Figure 4. Simulated microvibration signal in waterfall and contour plots in the frequency domain

The Fig. 4 and 5 illustrate the comparability of the RDM based simulated vibration signature and the measured microvibration signature of a specific reaction wheel on a test bed.

At a first glance, the modelled signature (which is not yet tuned to optimally fit the measurement) is already very similar to the measured one. The most relevant harmonics are present in the model and excite the same structural modes. Notably, the unbalance is estimated rather accurately, and the maximum disturbance is caused by the same harmonic order component crossing the same structural resonance at almost identical wheel speeds.

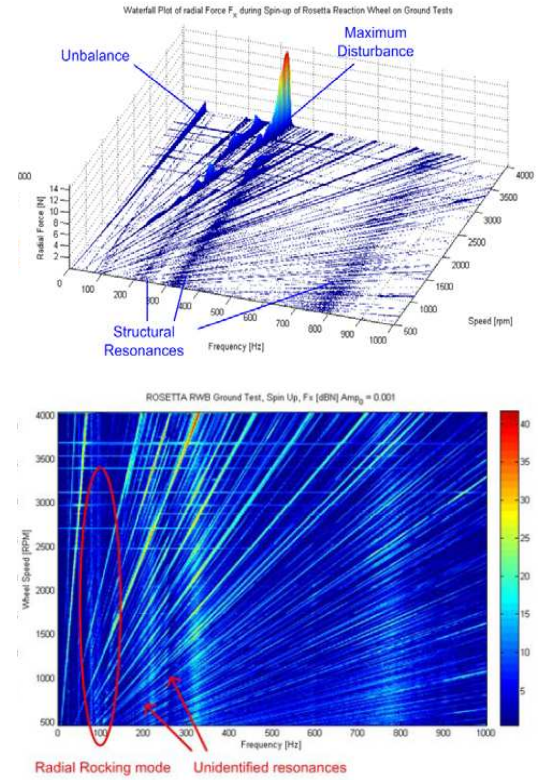


Figure 5. Measured vibration signature in waterfall and contour plots in frequency domain

However, there are still some aspects, which are not modelled accurately, compared to the measurement. The most evident differences are:

- Over-estimation of amplification by structural resonances at 220 Hz and 320 Hz
- Radial rocking wheel mode under-estimated
- Side band components and multiples of the basic harmonics are under-estimated, especially between 300 Hz and 800 Hz
- Harmonic orders > 30 are not included in the simulated signal

Basically, these differences may be explained by the application of non-optimized model parameters (a, b, q, ζ). The actual model approach however can be verified, when focusing on an individual harmonic order, for which a first optimization has been performed. Fig. 5 shows the extracted harmonic order H6.11, which corresponds to the ball pass frequency for an inner raceway imperfection.

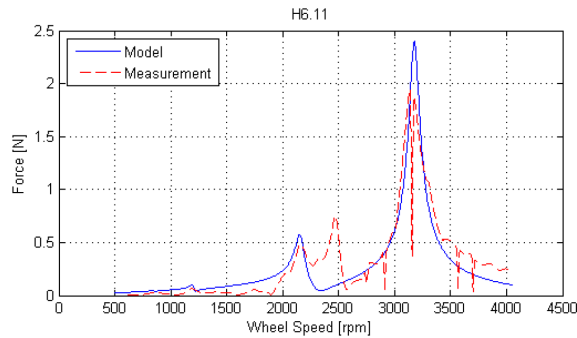


Figure 5. Simulated vs. measured amplitude – wheel speed correlation of the harmonic order H6.11

As can be seen from above diagram, the model can be fitted very well to the measurement in terms of peak location and peak height as well as overall evolution. This proves that the chosen model approach is capable of simulating the microvibration signature of a reaction wheel with adequate precision. The residual error with respect to the measurement are mainly due to distortions of the measurement signal (signal drop-outs at 2900 rpm, 3200 rpm, etc.) and a non-identified (and non-modelled) structural resonance passed at 2500 rpm.

3.4. Improving the RDM Simulation by Measurement based Parameter Fitting

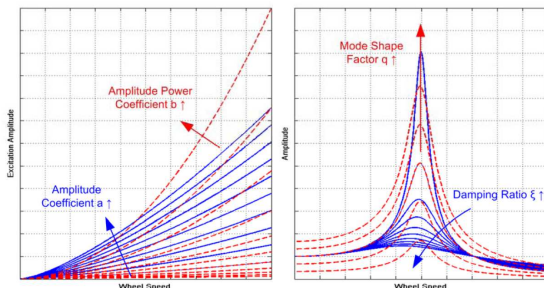


Figure 6. RDM parameter adjustment

RDM performance is mainly driven by the accurate definition of all RDM parameters, excitation parameters (amplitude coefficients) as well as structural parameters (resonance frequencies, damping ratio, modal participation factors). Fig. 6 illustrates the impact of the different RDM parameters on the model performance. The parameter a is a linear scaling factor to the induced force of a certain harmonic order. For order 1 (unbalance effect), this factor is identical to the unbalance parameter, depending on the rotor mass and the Centre of Gravity (CoG) offset. The parameter b is the exponent applied to the wheel speed. For the unbalance effect, this parameter is, in theory, equal to 2, but has been found to be slightly smaller with the studied examples. For the bearing harmonics, it is assumed to be 1 for the simulation results presented above. It might vary for different harmonics, though. From the simulated harmonic shown in Fig. 5, it might

be concluded that the estimated exponent is too small, such that some larger value might result in a better model correlation for this specific harmonic.

The factors q and ζ control the amplification of the signal as a certain harmonic excitation crosses a specific resonance. The peak magnitude might be adjusted by q , and the overall shape by ζ . It needs to be mentioned that there are cross-coupling effects between different eigen-modes. An iterative algorithm is therefore required to fit the structural parameters. Furthermore, the adjustment of the shape of a certain resonance peak via the damping ratio ζ should also be performed iteratively. In parallel to this, the estimation of the excitation parameters needs to be refined for each iteration step.

A prototype algorithm for the approach described above has been defined as part of the study. Its functionality could be proved using synthesised “dummy” signals for correlation.

4. ANALYSING MICROVIBRATION SIGNATURES FOR HEALTH-MONITORING OF SPACE MECHANISMS

4.1. General Approach

As discussed earlier on, the microvibration components related to specific mechanical effects caused by the bearings and the motor can be characterised by harmonic orders, which are amplified by the structural dynamics of the system. The amplitudes of these harmonics increase as the related mechanical effects (e.g. defects) become more dominant. This idea has been utilised for non-intrusive health monitoring of rolling element bearings in terrestrial applications over the last 25 years approximately [12]. The related methods (based on frequency analysis, wavelet analysis, or time domain analysis, [13]) have been adopted for routine checks and maintenance. However, examples for a similar implementation on launcher and spacecraft mechanisms are still very rare (e.g. done for the Space Shuttle Main Engine [14]).

Especially for rotation mechanisms, a health monitoring system might be required to detect:

- Amplitude changes for selected harmonic orders
- Small shifts of harmonic orders
- Small variations of resonances (in terms of eigen-frequencies and damping ratios)

The corresponding performance indicators might be obtained by iteratively fitting the RDM to a given measurement data set. For this purpose, it is first required to track and extract the selected harmonic order (e.g. bearing related or motor related) from appropriate measurement data for preparation of subsequent analyses. Several methods for this are already reported in literature. In the framework of the study, selected methods have been implemented and shall be briefly reviewed in the following paragraph

4.2. Order Analysis of Reaction Wheel Microvibration Signatures

For assessing the health conditions of a rotation mechanism, the relevant harmonics need to be identified and extracted from a measurement data set to derive the respective amplitude vs. wheel speed characteristics for further analysis.

Order Tracking

Techniques to identify harmonic orders from a given measurement are called order tracking methods and are already well known in the art [15]. For the analyses performed within this work so called phase synchronised order tracking [15], [16], [17].

The Fast Fourier Transform (FFT) based order tracking method is based on the principle of angular re-sampling. Hence, in a preparatory step the measured microvibration signal, which is typically acquired in the time domain, is re-sampled using information on the absolute (or relative) rotation angle of the rotor (see Fig. 3 and Fig. 4). After re-sampling, the measured signal is available with constant angular increments. Performing an FFT on the re-sampled signal provides the spectral estimation directly in the order domain. For the re-sampling routine, a tachometer pulse train with adequate temporal resolution is well suited to derive the instantaneous absolute/relative angle of rotation. Alternatively, this information may be obtained by numerical integration of the wheel speed signal. This method provides the spectral decomposition of the measurement signal as function of the wheel speed in the order domain.

In general, this method might require more computational effort than simply transforming a frequency spectrum into an order spectrum by an argument change using the wheel speed vector. It can, however, achieve higher resolution in the order domain (due to phase synchronisation) and, hence, it is capable of identifying harmonic orders in microvibration data very accurately. Alternatively, the following methods have been published in literature:

- Vold-Kalman filtering [15]
- Time variant Fourier transform order tracking

In a particular instance, a Coherent Phase Line Enhancer (CPLE) has been applied within the Space Shuttle programme, in order to monitor the health state of turbo pump bearings. A detailed description can be found in [14].

Statistical analysis tools like histograms applied on the order spectra can then be utilised to provide a listed output of the most crucial harmonic orders.

Order Extraction

The term order extraction covers methods to extract the amplitude vs. wheel speed characteristic of a certain harmonic order from a given measurement data set. This step is required to actually describe the mechanical effect a selected harmonic order is based

on. For this purpose, the order is exported from the wheel speed depending order spectrum, e.g. obtained by phase synchronised order tracking.

4.3. Health and Condition Monitoring Approach for Reaction Wheels

Two health-monitoring approaches or scenarios might be considered of on-board utilisation on spacecraft: either a “maintenance mode” or an “on-line health monitoring mode”.

The first, maintenance mode, implies that during in-orbit operation (or possibly also during ground testing) full spin-up and spin-down microvibration measurements of the wheels are acquired, e.g. after certain operation time intervals. Using these measurements, all relevant harmonics can be identified and the according amplitude-speed relations can be analysed. The related features, e.g. the RDM parameters can be extracted. The comparison of these values with reference values, e.g. acquired during or after a run-in procedure or at the begin of life might unveil some trend of misbehaviour due to degradation or wear.

The associated values may be compared to defined thresholds such that the operational performance can be judged with regard to its long-term stability and criticality. This approach may be also applied for resonance frequencies and damping ratios, as the structural integrity of the reaction wheel assembly and the interface structure can be assessed using these parameters.

Additionally, an on-line health monitoring approach might be considered for a use in reaction wheel equipped spacecraft. This can be applied by quasi-permanently acquiring a microvibration signal of a reaction wheel (sensors need to be very close to the corresponding wheel for reducing the influence of other sources as much as possible). Based on this, short-time Fourier transform on signal segments can be performed, and the corresponding spectra can be analysed concerning the amplitudes of certain harmonics by taking into account the instantaneous wheel speed. As certain harmonic amplitudes exceed certain thresholds (thresholds need to be dependent on the instantaneous wheel speed to take structural resonances into account), a warning or fault indicator can be transmitted to the on-board data handling system, or more specifically, to a Failure Detection, Isolation and Recovery (FDIR) system. Basically, if the anomalous behaviour persists or becomes worse, a more detailed inspection, for example by applying the maintenance mode, may be applied to judge the criticality of the mechanism performance.

However, a space mechanism health monitoring system based on this idea might require high computational effort and broad-band telemetry. To reduce the related communication requirements to a reasonable level, most of the necessary signal processing and, hence, data reduction should be performed on-board.

5. CONCLUSION

Using the modal superposition approach a generic reaction wheel microvibration model could be implemented and compared with measurement data acquired during the microvibration test campaign of a reaction wheel. The model is capable for accurate simulations of the microvibration signatures of a reaction wheel, by also taking into account the gyroscopic effect as well as the impact of the mounting interface.

Within the frame of this project algorithms have been developed to iteratively extract such parameters from measurement data. Particularly parameters characterising the regarded excitation sources also may be considered as measures for the mechanical conditions inside the mechanism (amount of local defects of the bearings, wear, etc.).

6. LESSONS LEARNED

The modal superposition approach is very well suited to simulate the dynamic behaviour of a reaction wheel mounted on an arbitrary interface.

By considering the problem piecewise linear (hence, discrete wheel speed input) the gyroscopic effect can be included in the transfer function, enhancing the model performance.

Model parameters and spectral estimation parameters need to be selected carefully as the model performance is very sensitive to variations.

Using iterative optimisation approaches the model can be fitted to a measurement data set of a real reaction wheel. The respective model parameters (especially excitation parameters) can be considered as measures for the mechanical conditions.

Algorithms for the automatic extraction of such parameters from measurements can be a reasonable basis for future health monitoring.

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